CyberSentinel AI

SOC INCIDENT REPORT

Incident ID	INC-764BF11D6948D241
Threat	Brute Force
Event time (UTC)	2026-09-01T02:00:00+00:00
Generated at	2026-10-07T18:56:38.717039+00:00
Severity / Status	High / Open
Confidence / Risk	65% heuristic / 70/100 heuristic

Decision support only. Confidence and risk are heuristic. Disruptive response actions require human approval. Unsupported Unicode is replaced with ? in this PDF; JSON preserves original text.

Incident Summary

Possible brute force on endpoint-024 at 2026-09-01T02:00:00+00:00. Isolation Forest identified an unusual feature pattern; analyst validation is required.

Threat Explanation

Telemetry is consistent with the brute force triage hypothesis. Authorized administration, testing, or workload changes may produce similar observations.

Technical Explanation

Model MODEL-C4BD9AD7BD129775 used eight numeric features with log1p scaling. Anomaly score 0.6089; decision function -0.0580. The candidate threat derives from numerical thresholds and message indicators, not dataset labels.

Business Impact

Potential service disruption, account misuse, or data exposure. Actual impact and scope are unconfirmed; verify asset role and data classification.

MITRE ATT&CK (Candidate Mappings)

T1110 - Brute Force: Candidate mapping for Brute Force; validate with additional telemetry. [mitre/brute_force.md#chunk-0]

Evidence

timestamp: 2026-09-01T02:00:00+00:00

source_ip: 10.20.0.34

destination_ip: 198.51.100.25

user: analyst_24

host: endpoint-024

event_type: Brute Force

message: Repeated authentication failures followed by a success

asset_criticality: medium

failed_attempts: 127.0

login_frequency: 111.0

bytes_sent: 198478.0

bytes_received: 2780911.0

request_rate: 68.0

unique_ports: 7.0

authentication_failures: 88.0

destination_count: 4.0

ground_truth: anomaly

event_id: EVT-596F7FAA4201B4A9

prediction: -1

anomaly_score: 0.6089257566522382

decision_function: -0.05800768572341075

confidence: 0.9765624999999996

reference_percentile: 0.99625

model_id: MODEL-C4BD9AD7BD129775

Attack Chain (Observed and Hypothesized)

1. Observed: Repeated authentication failures followed by a success
2. Hypothesis: Credential Access activity; not independently confirmed.
3. Next: correlate endpoint, identity, and network records to establish sequence.

Recommended Actions

1. Review account lockout and sign-in policy.
2. Enable phishing-resistant MFA.
3. Rate-limit authentication after verifying legitimate traffic.

Recommended Mitigation

1. Review account lockout and sign-in policy.
2. Enable phishing-resistant MFA.
3. Rate-limit authentication after verifying legitimate traffic.

SOC Checklist

1. Validate event time and collection source.
2. Confirm business authorization.
3. Preserve raw logs and evidence hashes.
4. Assess asset criticality and affected identities.
5. Document containment approval.
6. Assign investigation owner and escalation deadline.

Next Investigation Steps

1. Search surrounding 30 minutes for the same host, user, and source.
2. Correlate endpoint process ancestry and identity session records.

3. Check related hosts and destinations against authorized workflows.

4. Confirm candidate MITRE mapping with independent evidence.

Recovery

1. Verify clean backups and known-good configurations.

2. Restore in stages with the service owner; monitor health and indicators.

3. Rotate exposed credentials after approved containment when exposure is established.

Uncertainties

1. An anomaly alone does not prove malicious intent.

2. No endpoint process tree, packet capture, or corroborating identity timeline was supplied.

3. Attack chain stages are hypotheses requiring independent evidence.

Response Playbook

Triage

Confirm anomaly, asset owner, scope, and change window.

Owner: SOC analyst | Approval required: False

Validation: Document whether activity was authorized and record corroborating sources.

Preserve

Collect relevant logs and forensic artifacts with hashes and timestamps.

Owner: Incident responder | Approval required: False

Validation: Evidence manifest and chain of custody are complete.

Contain

Review account lockout and sign-in policy. Approve any account, host, or network restriction before execution.

Owner: Incident commander | Approval required: True

Validation: Containment scope is approved and business impact assessed.

Eradicate

Rate-limit authentication after verifying legitimate traffic.

Owner: Platform owner | Approval required: True

Validation: Root cause has been addressed and telemetry reviewed.

Recover

Restore validated services, check integrity, and monitor for recurrence.

Owner: Service owner | Approval required: True

Validation: Service tests pass and recovery is approved.

Learn

Record lessons, update detection logic, and review response timing.

Owner: SOC lead | Approval required: False

Validation: Review is recorded with accountable owners.

Retrieved Sources

mitre/brute_force.md#chunk-0 - Brute Force | similarity=0.6310

- https://attack.mitre.org/techniques/T1110/

mitre/credential_dumping.md#chunk-0 - OS Credential Dumping | similarity=0.5034

- https://attack.mitre.org/techniques/T1003/

mitre/valid_accounts.md#chunk-0 - Valid Accounts | similarity=0.4972

- https://attack.mitre.org/techniques/T1078/

mitre/rdp.md#chunk-0 - Remote Desktop Protocol | similarity=0.4951

- https://attack.mitre.org/techniques/T1021/001/

mitre/defense_evasion.md#chunk-0 - Indicator Removal | similarity=0.4831

- https://attack.mitre.org/techniques/T1070/

Provenance

analysis_provider: local

requested_provider: local

fallback_reason:

retrieval_mode: sentence_transformer / faiss

kb_fingerprint: aa8c954cff09c6a1375ebccfdd5a2e6935c129bc2088452875b0c45465a19e9a

model_id: MODEL-C4BD9AD7BD129775

pipeline_version: 1.0.0

confidence_definition: Uncalibrated analyst heuristic; not a probability.

risk_definition: Heuristic prioritization on 0-100 scale; not measured business loss.

Agent Execution Trace

Detection Agent: 0.03 ms (completed)

Threat Intelligence Agent: 24.17 ms (completed)

Analysis Agent: 0.38 ms (completed)

Response Agent: 0.08 ms (completed)

Report Agent: 0.06 ms (completed)